

## **Reflection Paper on the Stuxnet Virus Incident through CIA Triage**

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## **Introduction**

The documentary “Zero Day” is a movie about the Stuxnet virus and how it was created primarily by the United States to target the Natanz nuclear facility in Iran and their Siemens Programmable Logic Controllers. Their motive for targeting this nuclear facility involved preventing them from creating nuclear weapons to ensure the safety of citizens in many countries from harm.

We can use the confidentiality, integrity, and availability (CIA) triage to identify ways the Stuxnet virus could have been better contained from the point of view of security engineers at the Natanz Iranian Nuclear Facility. The following details will provide an analysis of how the Stuxnet virus could have been prevented or mitigated.

## **CIA Triad**

### **Confidentiality**

Confidentiality refers to the protection of information from unauthorized access, and the data of an organization is kept private. Firstly, the facility should not have allowed the released photographs within the facility. These photographs were taken during a tour for the Iranian president, and they leaked information about computer screens and the centrifuge models that the nuclear facility used. The photos helped creators of the Stuxnet virus to create code to specifically target those centrifuges and disrupt their functions.

Access control policies for digital media, such as USB drives, could have also been implemented to prevent the Stuxnet virus from infecting the facility systems. Blocking all unauthorized devices, disabling autorun and autoplay features in operating systems, and ensuring read-only access (Arrospide) are controls that could have helped prevent Stuxnet from affecting their facility systems.

Lastly, the facility could have ensured that network segmentation was used wherever applicable. For example, the control systems of the centrifuges should be isolated from the administrative networks and any other unrelated networks. There is a possibility that a USB drive infected with the Stuxnet virus could have been initiated in a far section of the network and automatically spread to the centrifuge systems.

### **Integrity**

Integrity refers to ensuring that data is not unintentionally changed or erased. Stuxnet successfully corrupted the data that the centrifuges were sending back to the system controls. The operator screens showed that the centrifuges were spinning at the correct speeds when they were being sped up and slowed down, causing them to fail.

Among other forms of defence, an intrusion detection system or real-time monitoring could have been used to monitor traffic. Human operators would be able to see data from the centrifuges on the operator screens and identify any suspicious activity that the Stuxnet virus was performing.

Additionally, an independent monitor that would have been a secondary, offline method to see the spin of the centrifuges removes the possibility of altered measurements from displaying (How Nozomi Networks could have mitigated the Stuxnet attack). The automated systems could

have been consistently supplemented with manual checks, given the workers provided to perform them.

### **Availability**

Availability guarantees that authorized users can access systems and data as required. Stuxnet successfully prevented the availability of the centrifuges by causing them to fail. Some actions of containment would be to implement a patch management policy and a manual emergency shutdown procedure.

The patch management policy would deal with the zero-day vulnerabilities that the Stuxnet virus took advantage of by updating the security of the systems (Insights, 2024). The emergency shutdown procedures would aim to manually stop the centrifuges from spinning without the event of an explosion, crash, or any physical damage.

Additional ways that the security engineers could have contained the virus would be to harden the systems and stress the air gap enforcement. System hardening would involve ensuring that the bare minimum of software and services would be met and that all supplementary or unnecessary software would be removed. This would also ensure that applications are inspected and approved before they can run. Stressing the air gap enforcement would mean thoroughly scanning external devices such as USB drives before introducing them to the facility systems.

### **Conclusion**

From the view of a security engineer at the Natanz Iranian Nuclear Facility, containing the Stuxnet virus proved a challenge that required a multi-layered defence that would closely follow the CIA triage. By implementing strict network segmentation, access controls, manual consistent validation, intrusion detection, hardening systems, and stressing air gap controls, the facility could have significantly reduced the impact of the attack. With these security actions, the nuclear facility would grow and continue to have production without delay.

## References

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